Quantum Physics, 2026 Fall, BME
Lecture 05 - Perturbation Theory in Quantum Mechanics

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1 When do we use perturbation theory?

1. It often happens that (i) there is a quantum system we understand, but (ii) we apply a small modification to it, and (iii) we want to know the small changes that are induced by this small modification.

2. By the phrase 'a quantum system we understand', I mean a quantum system whose Hamiltonian (an $n \times n$ Hermitian matrix H^0) is known, and also the eigenvectors $|\psi_j\rangle$ and eigenvalues E_j of H^0 are known.

3. By 'a small modification' to the quantum system, I mean a 'small' correction H' (also an $n \times n$ Hermitian matrix) to the original Hamiltonian, so that the Hamiltonian is changed to $H = H^0 + H'$. The term H' is called the perturbation.

4. If the perturbation H' is 'small', then the eigenvectors $|\psi_j^{\text{pert}}\rangle$ and eigenvalues E_j^{pert} of H are only 'slightly' different from those of H^0, and we want to describe these slight differences.

5. Example: Electron spin in a slightly tilted magnetic field.

   (i) Consider an electron spin in a magnetic field $H^0 = \frac{1}{2} g \mu_B B_z \sigma_z$, with $B_z = 1$ T. The eigenvectors and eigenvalues of this H^0 are: $E_1 = -\frac{1}{2} g \mu_B B_z$, $|\psi_1\rangle = |\downarrow\rangle$, $E_2 = \frac{1}{2} g \mu_B B_z$, $|\psi_2\rangle = |\uparrow\rangle$. The Hamiltonian H^0 is called the 'unperturbed' Hamiltonian,

whereas the eigenvalues and eigenvectors are called the 'unperturbed' eigenvalues and eigenvectors.

(ii) Apply a small magnetic field $B_x = 0.1$ T along x, such that $H = H^0 + H'$ with $H' = \frac{1}{2} g \mu_B B_x \sigma_x$.

(iii) What are the modified eigenvalues and eigenvectors, i.e., those of H? They are called 'perturbed' eigenvalues and eigenvectors.

6. Further (abstract) examples:

$$H^0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \; H' = \begin{pmatrix} 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 \end{pmatrix} \tag{1}$$

$$H^0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \; H' = \begin{pmatrix} 0 & 0 & 0.1 \\ 0 & 0 & 0.1 \\ 0.1 & 0.1 & 0 \end{pmatrix} \tag{2}$$

7. We think of the perturbation H' as a single-parameter family of perturbations, $H' = \nu V$, where V is a Hermitian matrix, and ν is a real number we call the 'perturbation strength'.

8. In perturbation theory, we express the perturbed eigenvalues and eigenstates as a power series in the perturbation strength ν, and we truncate the power series at a finite order (often at first or second order).

9. When does this approach provide an accurate description of the perturbed eigenvalues and eigenstates? If ν is 'small enough'. (We will not study this condition rigorously.)

10. Remark: Perturbation theory described here is linear algebra for Hermitian matrices, with quantum physics applications. The same theory is also valid for perturbation of real symmetric matrices, and hence applicable to physics, for, e.g., dynamical matrices of classical mechanical systems.

2 Nondegenerate perturbation theory

2.1 General formalism

1. Here we assume that the jth eigenvalue E_j of the unperturbed Hamiltonian H^0 is non-degenerate:

$$E_1 \leq E_2 \cdots \leq E_{j-1} < E_j < E_{j+1} \leq E_{j+2} \leq \cdots \leq E_n. \tag{3}$$

That is, there are nonzero energy gaps $\Delta_+ := E_{j+1} - E_j$ and $\Delta_- := E_j - E_{j-1}$ separating E_j from its neighbors.

2. As a rule of thumb, the approximate formulas shown in this lecture are valid if all matrix elements $\nu \langle \psi_k | V | \psi_l \rangle \equiv \langle \psi_k | H' | \psi_l \rangle$ of the perturbation between the unperturbed eigenvectors are much smaller than both gaps Δ_+ and Δ_-.

3. The change of the jth energy eigenvalue E_j due to the perturbation is expressed as a power series in ν as follows:

$$E_j^{\text{pert}} = E_j + E_j^{(1)} + E_j^{(2)} + \dots \tag{4}$$

4. Formula for the first-order correction of the jth energy eigenvalue:

$$E_j^{(1)} = \nu \langle \psi_j | V | \psi_j \rangle . \tag{5}$$

5. Formula for the second-order correction of the jth energy eigenvalue:

$$E_j^{(2)} = \nu^2 \sum_{k \neq j} \frac{|\langle \psi_k | V | \psi_j \rangle|^2}{E_j - E_k}. \tag{6}$$

6. Formula for the first-order correction of the $j$the eigenvector:

$$|\psi_j^{\text{pert}}\rangle = |\psi_j\rangle + |\psi_j^{(1)}\rangle , \tag{7}$$

where

$$|\psi_j^{(1)}\rangle = \nu \sum_{k \neq j} |\psi_k\rangle \frac{\langle \psi_k | V | \psi_j \rangle}{E_j - E_k} \tag{8}$$

7. We do not derive the above results here. For derivation, see section 6.1. 'Non-degenerate perturbation theory' of the book Griffiths: 'Introduction to Quantum Mechanics'. The eigenvalue correction formulas can also be obtained straightforwardly from the results of section 3 of this lecture.

2.2 Example: Electron spin in a slightly tilted magnetic field

1. As a solution for 'Electron spin in a slightly tilted magnetic field' example above (section 1, item 5), we express the eigenvalue corrections up to second order and the eigenvector corrections up to first order:

$$E_1^{\text{pert}} = -\frac{1}{2} g \mu_B B_z - \frac{1}{4} g \mu_B \frac{B_x^2}{B_z}, \tag{9a}$$

$$E_2^{\text{pert}} = \frac{1}{2} g \mu_B B_z + \frac{1}{4} g \mu_B \frac{B_x^2}{B_z}, \tag{9b}$$

$$|\psi_1^{\text{pert}}\rangle = |\downarrow\rangle - \frac{B_x}{2B_z} |\uparrow\rangle , \tag{9c}$$

$$|\psi_2^{\text{pert}}\rangle = |\uparrow\rangle + \frac{B_x}{2B_z} |\downarrow\rangle . \tag{9d}$$

2. Remark 1: This example is so simple that exact diagonalization is possible, no need to use perturbation theory. For example, the exact eigenvalues are

$$E_1^{\text{pert}} = -\frac{1}{2} g \mu_B \sqrt{B_z^2 + B_x^2}, \tag{10a}$$

$$E_2^{\text{pert}} = \frac{1}{2} g \mu_B \sqrt{B_z^2 + B_x^2}. \tag{10b}$$

This enables comparison of exact and perturbative solutions, see Figure 1.

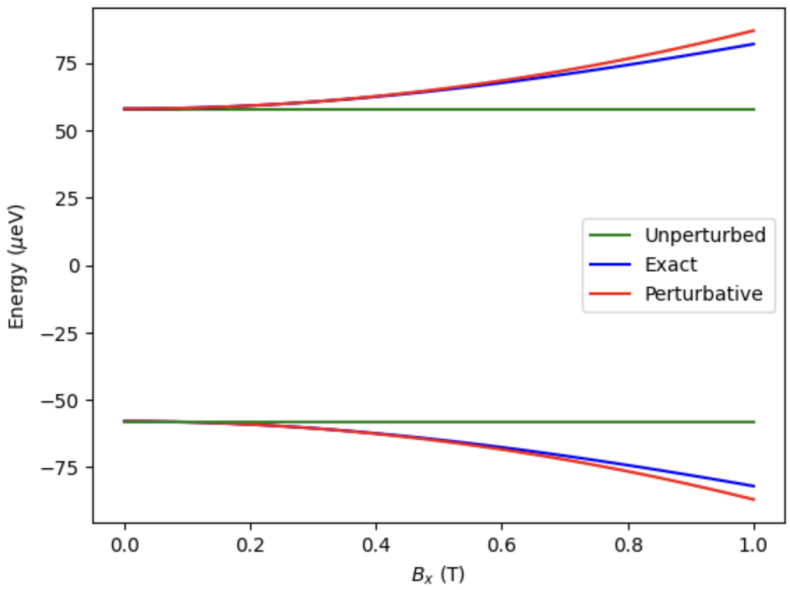


Figure 1: Comparison of exact (blue, Eq. (10)) and perturbative (red, Eqs. (9a) and (9b)) energy spectrum of a spin placed in a magnetic field, with $B_z = 1$ T.

3. Remark 2: Note the effect of 'level repulsion': the second-order correction of the jth eigenvalue due to the interaction with lower eigenvalues ($j' < j$) push the eigenvalue upwards, whereas the correction of the jth eigenvalue with higher eigenvales ($j' > j$) push the eigenvalue downwards.

4. Remark 3: The perturbed wavefunctions resulting from finite-order perturbation theory are not orthonormal. They are approximately orthonormal. E.g., the first-order perturbed states above are approximately orthonormal in the following sense:

$$\langle \psi_i^{\text{pert}} | \psi_j^{\text{pert}} \rangle = \delta_{ij} + O\left(\frac{B_x^2}{B_z^2}\right). \tag{11}$$

2.3 Graphical representation of perturbation theory

1. The matrix $H = H^0 + H'$ is represented as a graph. Nodes: unperturbed eigenvectors. Edges: nonzero matrix elements between unperturbed eigenvectors. (Loop-type edges are possible!) Nodes are drawn as short vertical lines with a horizontal position representing the corresponding energy. Edges are drawn as two-head arrows.

2. Corrections of the jth energy eigenvalue in nth order ($n \in \mathbb{N}^+$) are represented as n-long cyclic directed walks starting and ending on the jth node.

3. Corrections of the jth eigenvector in nth order are represented as n-long directed walks starting at the $j$the node.

4. Exercise: Draw the graph representing the 'Electron spin in a slightly tilted magnetic field' example, section 1, item 5. Draw the cycles representing the first-order and second-order energy corrections of the ground state.

5. Exercise: Draw the graph corresponding to Eq. (1). Draw the cycles representing the first-order and second-order corrections of the ground state.

3 Quasidegenerate perturbation theory

1. Example: Eq. (2).

2. Task: Consider the case when H^0 has two sets of eigenvalue-eigenvector pairs, A and B, such that eigenvalues in A are well separated from eigenvalues in B. Express the change of the eigenvectors or eigenvalues in A caused by the perturbation H'.

3. Terminology: A is called the *relevant subspace*, B is called the *irrelevant subspace*.

4. Idea for solution: if the perturbation H' is 'small', then we can hope to find a 'small' unitary transformation U such that

$$\tilde{H} = UHU^\dagger \tag{12}$$

is block-diagonal. ('Block-diagonal' refers to the block structure defined by A and B.)

5. This is useful: e.g., if the dimension of the Hilbert space is $n = 100$, but the dimension of the relevant subspace is $n_A = 2$, then this approach yields analytical results for the relevant eigenvalues and eigenvectors.

6. Claim: the above idea works, and provides formulas for the transformation U and for the relevant block of $\tilde{H}$.

   Step 1. The desired unitary matrix is expressed as

$$U = e^{-S}, \tag{13}$$

   where S is a block-off-diagonal anti-Hermitian matrix ($S^\dagger = -S$).

   Step 2. The anti-Hermitian matrix S is searched as a power series in the strength of the perturbation,

$$S = S^{(1)} + S^{(2)} + S^{(3)} + \dots \tag{14}$$

   Step 3. Substituting Eq. (14) into the condition that $\tilde{H}$ is block-off-diagonal yields order-by-order linear inhomogeneous equations for the components $S^{(1)}$, $S^{(2)}$, $S^{(3)}$, ... of the anti-Hermitian matrix, which can be solved. For example, the matrix

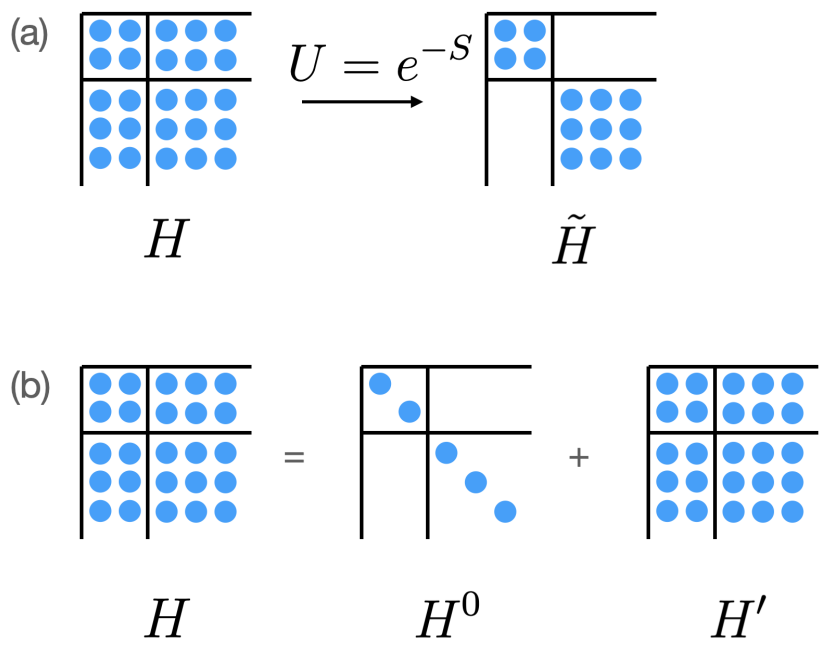


Figure 2: Quasidegenerate perturbation theory is based on a 'small' unitary transformation $U = e^{-S}$ that renders the Hamiltonian H to a block-diagonal form $\tilde{H}$, as shown in panel (a). The method exploits that the unperturbed Hamiltonian H^0 is diagonal, see panel (b), and the perturbation H' is weak.

elements of the top right block of $S^{(1)}$ and $S^{(2)}$ are expressed as (with $m \in A$ and $l \in B$):

$$S^{(1)}_{ml} = -\frac{H'_{ml}}{E_m - E_l}, \tag{15a}$$

$$S^{(2)}_{ml} = \frac{1}{E_m - E_l}\left[\sum_{m' \in A} \frac{H'_{mm'}H'_{m'l}}{E_{m'} - E_l} - \sum_{l' \in B} \frac{H'_{ml'}H'_{l'l}}{E_m - E_{l'}}\right]. \tag{15b}$$

Step 4. Using the solution for S, the transformed Hamiltonian can be expressed order-by-order from Eq. (12):

$$\tilde{H} = e^{-S}He^{S} = H^{(0)} + H^{(1)} + H^{(2)} + \ldots. \tag{16}$$

In fact, we care about the relevant block of $\tilde{H} = e^{-S}He^{S}$, whose matrix elements up to second order are expressed as (with $m, m' \in A$):

$$H^{(0)} = H^0_{mm'}, \tag{17a}$$

$$H^{(1)} = H'_{mm'}, \tag{17b}$$

$$H^{(2)} = \frac{1}{2}\sum_{l \in B} H'_{ml}H'_{lm'}\left[\frac{1}{E_m - E_l} + \frac{1}{E_{m'} - E_l}\right]. \tag{17c}$$

Step 5. Diagonalize the $n_A \times n_A$ block of the transformed Hamiltonian to obtain the n_A perturbed eigenvalues $\tilde{E}_j$ and perturbed eigenvectors $|\tilde{\psi}_j\rangle$ of the relevant block.

Step 6. The perturbed eigenvalue $E_j^{\rm pert}$ of the relevant subspace is obtained as $\tilde{E}_j$. However, the perturbed eigenvectors in the original basis (eigenbasis of H_0) need to be expressed up the desired order using the inverse $U^{-1} = U^\dagger$ of the transformation U:

$$|\psi_j^{\rm pert}\rangle = U^\dagger \, |\tilde{\psi}_j\rangle \, . \tag{18}$$

7. Remark: If we care about perturbed eigenvalues only, then we should follow a simple workflow: (i) evaluate the relevant block of the transformed Hamiltonian up to the desired order using Eq. (17), and (ii) compute the eigenvalues of that relevant block.

   If we also care about the perturbed energy eigenvectors, then we need to use Eqs. (15) and (18) as well.

8. Remark: *Degenerate perturbation theory* is the special case when the relevant eigenvalues of H_0 are equal.

9. Remark: Quasidegenerate or degenerate perturbation theory has a few other names, e.g., *Schrieffer-Wolff transformation* or *Löwdin partitioning*.

10. Quasidegenerate perturbation theory and the derivation leading to Eqs. (15) and (17) is nicely described in the 5-page Appendix B of the book *Spin-orbit Coupling Effects in Two-Dimensional Electron and Hole Systems* by Roland Winkler.